

ELITE MATHEMATICS

Pearson Edexcel International GCSE

Linear 4MA1

Adaptive Classified

OL-T19 · Factorisation

With Answers

Questions with full worked solutions

■ 38 pages

FAMILY MAP

Factorising

F01	Factorise a non-monic or special quadratic	7 Q 21 marks
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F02	Extract an algebraic common factor	8 Q 22 marks
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F03	Factorise a monic quadratic	3 Q 6 marks
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Working Unit 05

4MA1-1H Q6

ORIGINAL QUESTION

3 marks

(b) (i) Factorise $y^2 - 11y + 30$

(2)

(ii) Hence solve $y^2 - 11y + 30 = 0$

(1)

Worked Solution 05

Factorise, then solve

UNIT 1
3 marks**Factorise**

$$y^2 - 11y + 30 = (y - 5)(y - 6).$$

Use the zero-product rule

$$(y - 5)(y - 6) = 0 \Rightarrow y = 5 \text{ or } y = 6.$$

Final answer

$$(y - 5)(y - 6), \quad y = 5 \text{ or } 6$$

Dependency check The 'Hence' solution depends on the factorisation and therefore remains in the same working unit.

Question 08

4MA1-1H Q12

ORIGINAL QUESTION

2 marks

12 (a) Factorise fully $6y^2 - 5y - 4$

(2)

Worked Solution 08

Factorise a quadratic

MODULAR UNIT 1

2 marks

Factorise

$$6y^2 - 5y - 4 = (2y + 1)(3y - 4).$$

Final answer

$$(2y + 1)(3y - 4)$$

Question 04

4MA1-1H Q2 (a)

ORIGINAL QUESTION UNIT 1

2 marks

2 (a) Factorise fully $12pq - 18p$
(2)

Worked solution 04

Factorise fully

MODULAR UNIT 1

2 marks

(a) – Factorise fully

Take out the common factor

$$12pq - 18p = 6p(2q - 3).$$

Final answer

$$6p(2q - 3)$$

Question 04

4MA1-1H Q6

ORIGINAL QUESTION

2 marks

(c) Factorise fully $16a^2b^3 + 20a^3b$

(2)

Worked Solution 04

Common-factor factorisation

MODULAR UNIT 1

2 marks

Take the greatest common factor

$$16a^2b^3 + 20a^3b = 4a^2b(4b^2 + 5a).$$

Final answer

$$4a^2b(4b^2 + 5a)$$

Question 07

4MA1-1H Q14

ORIGINAL QUESTION

3 marks

14 Factorise fully $50g^2 - 18$

Worked Solution 07

Difference of two squares

MODULAR UNIT 1

3 marks

Take out 2

$$50g^2 - 18 = 2(25g^2 - 9).$$

Factor the difference

$$25g^2 - 9 = (5g + 3)(5g - 3).$$

Final answer

$$2(5g + 3)(5g - 3)$$

Original question 07

4MA1-1H Q11 Q11(a)

ORIGINAL QUESTION**2 marks****11** (a) Factorise $9x^2 - 4y^2$

(2)

Worked solution

Family: Factorise a non-monic or special quadratic • Variant: Difference of two squares

METHOD / WORKING

MODULAR UNIT 1 • JUNE 2022 DETAILED SOLUTIONS
Worked solution 07 MODULAR UNIT 1
Chapter: Factorising 2 marks
Q11(a) – Chapter: Factorising
Family: Factorise a non-monic or special quadratic • Variant: Difference of two squares
Method
$9x^2 - 4y^2 = (3x)^2 - (2y)^2$.
Why: Recognise a difference of two squares.
Method
Therefore $9x^2 - 4y^2 = (3x + 2y)(3x - 2y)$.
Why: Use $a^2 - b^2 = (a + b)(a - b)$.
Final answer
factorisation: $(3x + 2y)(3x - 2y)$

FINAL ANSWER

Original question 19

4MA1-2H Q1 Q1(b)

ORIGINAL QUESTION**2 marks**

(b) Factorise fully $15b^5c - 35b^3c^9$

(2)

Worked solution

Family: Extract an algebraic common factor • Variant: Greatest monomial common factor

METHOD / WORKING

MODULAR UNIT 1 • JANUARY 2022 DETAILED SOLUTIONS
Worked solution 19 MODULAR UNIT 1
Chapter: Factorising 2 marks
Q1(b) – Chapter: Factorising
Family: Extract an algebraic common factor • Variant: Greatest monomial common factor
Method
$15b^5c - 35b^3c^9 = 5b^3c(3b^2 - 7c^8).$
Why: Take out the greatest common factor $5b^3c$.
Method
Expanding the bracket checks the original expression.
Why: Verify both terms after factorising.
Final answer
factorisation: $5b^3c(3b^2 - 7c^8)$

FINAL ANSWER

(b) Factorise fully $5y + 20y^2$

(2)

Worked solution

January 2020 | Question 7 | 2 marks

Family: Extract an algebraic common factor | Variant: Greatest monomial common factor

METHOD / WORKING

1. Take the greatest common factor $5y$.2. $5y + 20y^2 = 5y(1 + 4y)$.

FINAL ANSWER

 $5y(1+4y)$

5 (a) Factorise fully $15y^4 + 20uy^3$

(2)

Worked solution

January 2021 | Question 5 | 2 marks

Family: Extract an algebraic common factor | Variant: Greatest monomial common factor

METHOD / WORKING

1. Factor out the highest common factor $5y^3$.
2. The fully factorised form is $5y^3(3y + 4u)$.

FINAL ANSWER

$$5y^3(3y + 4u)$$

(b) (i) Factorise $x^2 + 5x - 36$

(2)

(ii) Hence, solve $x^2 + 5x - 36 = 0$

(1)

Worked solution

January 2021 | Question 9 | 2 marks

Family: Factorise a monic quadratic | Variant: Negative constant monic quadratic

METHOD / WORKING

1. Find two numbers with product -36 and sum 5: 9 and -4.

2. Therefore $x^2 + 5x - 36 = (x + 9)(x - 4)$.

FINAL ANSWER

 $(x+9)(x-4)$

9 (a) Factorise fully $25a^4c^7d + 45a^9c^3h$

(b) Solve $(2x + 5)^2 = (2x + 3)(2x - 1)$

.....
(2)

$x =$
(3)

Worked solution

June 2021 | Question 9 | 2 marks

Family: Extract an algebraic common factor | Variant: Greatest monomial common factor

METHOD / WORKING

1. Take the common factor $5a^3c^3$ from both terms.2. The remaining bracket is $5c^3d + 9a^3h$.

FINAL ANSWER

$$5a^3c^3(5c^3d + 9a^3h)$$

9 (i) Factorise $x^2 + 2x - 24$

(2)

(ii) Hence solve $x^2 + 2x - 24 = 0$

(1)

Worked solution

June 2021 | Question 9 | 2 marks

Family: Factorise a monic quadratic | Variant: Positive constant monic quadratic

METHOD / WORKING

1. Find two numbers with product -24 and sum 2: 6 and -4.

2. Therefore $x^2 + 2x - 24 = (x + 6)(x - 4)$.

FINAL ANSWER

 $(x+6)(x-4)$

17 (a) Factorise $6y^2 - y - 5$

(2)

Worked solution

November 2021 | Question 17 | 2 marks

Family : Factorise a non – monic special quadratic | Variant : Non – monic quadratic $ax^2 + bx + c$

METHOD / WORKING

1. (a)

$$2. 6y^2 - y - 5 = (6y + 5)(y - 1)$$

FINAL ANSWER

$$(6y + 5)(y - 1), f = \frac{8w - 3}{w + 2}, 4(x - 1)^2 + 3$$

(b) (i) Factorise $x^2 - 5x - 36$

(2)

(ii) Hence solve $x^2 - 5x - 36 = 0$

(1)

Worked solution

November 2021 | Question 8 | 2 marks

Family: Factorise a monic quadratic | Variant: Negative constant monic quadratic

METHOD / WORKING

1. (b)(i)

2. $x^2 - 5x - 36 = (x - 9)(x + 4)$

FINAL ANSWER

1, $(x-9)(x+4)$, $x=9$ or $x=-4$

(b) (i) Factorise $x^2 - 5x - 36$

(2)

(ii) Hence solve $x^2 - 5x - 36 = 0$

(1)

Worked solution

November 2021 | Question 8 | 1 marks

Family: Factorise a non-monic or special quadratic | Variant: Factorise as the first part of a factor-then-solve chain

METHOD / WORKING

1. (b)(ii) $x = 9$ or $x = -4$.2. Answer: 1, $(x - 9)(x + 4)$, $x = 9$ or $x = -4$

FINAL ANSWER

1, $(x-9)(x+4)$, $x=9$ or $x=-4$

- 7 (a) Solve $5(4 - x) = 7 - 3x$
Show clear algebraic working.

$$x = \dots\dots\dots (3)$$

- (b) Factorise fully $16m^3g^3 + 24m^2g^5$

$$\dots\dots\dots (2)$$

- (c) (i) Factorise $y^2 - 2y - 48$

$$\dots\dots\dots (2)$$

- (ii) Hence, solve $y^2 - 2y - 48 = 0$

$$\dots\dots\dots (1)$$

(Total for Question 7 is 8 marks)

Answer reference | May-Nov 2020 | Question 7

Family: Factorise a non-monic or special quadratic • Variant: Factorise as the first part of a factor-then-solve chain

7	(a)	$20 - 5x (= 7 - 3x)$		3	M1 for expansion of bracket
		E.g. $20 - 7 = -3x + 5x$ or $-5x + 3x = 7 - 20$			M1 ft from a 4-term equation for a correct process of isolating terms in x on one side of the equation and numbers on the other side
			6.5 oe		A1 dep on M1 awarded and from correct working
	(b)			2	M1 for any correct partial factorisation with at least 2 factors, one of which must be a letter or the correct common factor with no more than 1 error inside the bracket
			$8m^2 g^3 (2m + 3g^2)$		A1
	(c)(i)	$(y \pm 6)(y \pm 8)$		2	M1
			$(y - 8)(y + 6)$		A1
	(c)(ii)		8, -6	1	B1 must ft from their factors in (c)(i)
Total 8 marks					

Worked solution

May-Nov 2020 | Question 7 | 8 marks

Family: Factorise a non-monic or special quadratic | Variant: Factorise as the first part of a factor-then-solve chain

METHOD / WORKING

Worked solution
May-Nov 2020 Question 7 8 marks
Family: Factorise a non-monic or special quadratic Variant: Factorise as the first part of a factor-then-solve chain
Solution
(a)
$5(4-x)=7-3x$
$20-5x=7-3x$
$13=2x$
$x=(13)/(2)$
(b)
$16m^3g^3 + 24m^2g^5 = 8m^2g^3(2m + 3g^2)$
(c)(i)
$y^2 - 2y - 48 = (y - 8)(y + 6)$
(c)(ii) $y=8$ or $y=-6$.
Answer: $x = (13)/(2), 8m^2g^3(2m + 3g^2), (y - 8)(y + 6), y = 8$ or $y = -6$

FINAL ANSWER

10 (a) Simplify $(6m)^0$

(1)

(b) Solve $5y + 20 < 7y + 1$

(2)

(c) Factorise fully $15w^2x^5 + 25w^3x^2$

(2)

(d) Simplify fully $\frac{36a^5c^7}{12a^2c^3}$

(2)

(Total for Question 10 is 7 marks)

Worked solution

November 2025 | Question 10 | 7 marks

Family: Extract an algebraic common factor | Variant: Common factor followed by a second factorisation

METHOD / WORKING

Worked solution
November 2025 Question 10 7 marks
Family: Extract an algebraic common factor Variant: Common factor followed by a second factorisation
Solution
(a) $(6m)^0=1$.
(b)
$5y+20<7y+1$
$19<2y$
$y>(19)/(2)$
(c)
$15w^2x^5 + 25w^3x^2 = 5w^2x^2(3x^3 + 5w)$
(d)
$(36a^5c^7)/(12a^2c^3) = 3a^3c^4$
Answer: $1, y > (19)/(2), 5w^2x^2(3x^3 + 5w), 3a^3c^4$
PROVISIONAL WORKED SOLUTION — NOVEMBER 2025 QP-ONLY EXCEPTION

FINAL ANSWER

7 (i) Factorise $x^2 + 2x - 48$

(2)

(ii) Hence, solve $x^2 + 2x - 48 = 0$

(1)

(Total for Question 7 is 3 marks)

Worked solution

November 2025 | Question 7 | 3 marks

Family: Factorise a non-monic or special quadratic | Variant: Factorise as the first part of a factor-then-solve chain

METHOD / WORKING

Worked solution
November 2025 Question 7 3 marks
Family: Factorise a non-monic or special quadratic Variant: Factorise as the first part of a factor-then-solve chain
Solution
$x^2 + 2x - 48 = (x + 8)(x - 6)$
So
$(x+8)(x-6)=0$
Answer: $x=-8$ or $x=6$
PROVISIONAL WORKED SOLUTION — NOVEMBER 2025 QP-ONLY EXCEPTION

FINAL ANSWER